

RNA-seq Bioinformatics Project

Early Detection Biomarkers for Infection Using RNA-seq Transcriptomics

A complete bioinformatics pipeline to discover molecular signatures distinguishing healthy individuals from infected patients — enabling early detection before clinical symptoms appear.

10,346

Genes

397

Samples

4

Infection Stages

20

Biomarkers

Project Details

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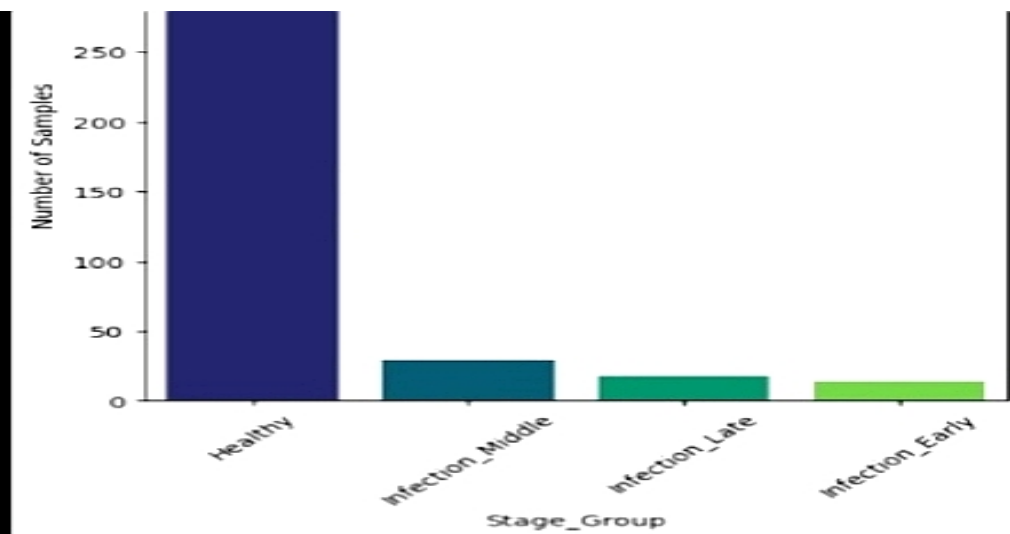
Type: Research Project

Tools Used: Python, T-test, PCA, Heatmap, Venn Diagram

Dataset: Diabetes & Infection RNA-seq Dataset (10,346 genes × 397 samples)

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Result 1: Dataset Overview — Sample Distribution per Stage



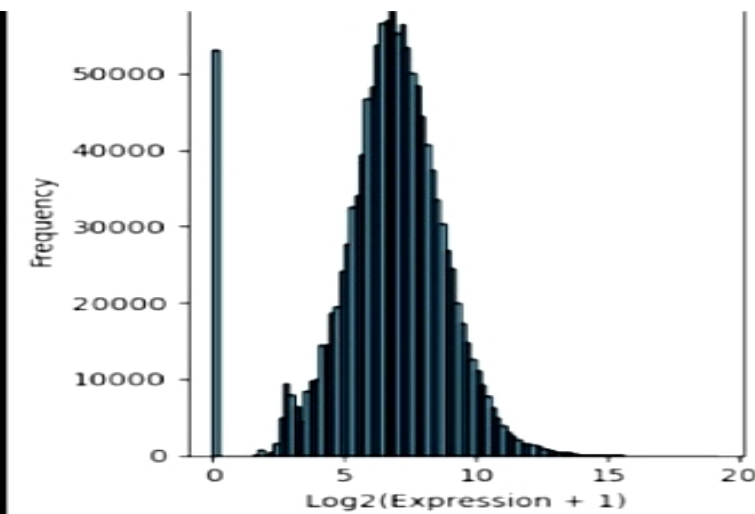
Commentary

Key Observation: The dataset is heavily imbalanced. The Healthy group dominates with ~336 samples; all three infection stages combined total only ~61 samples (Early ~14, Middle ~29, Late ~18).

Why it matters: This reflects reality — healthy patients outnumber infected ones. The T-test must compare these unequal groups carefully. The large Healthy baseline gives strong statistical power for defining 'normal' expression.

Conclusion: The Healthy group is our robust reference baseline. Infection_Early is the smallest and hardest to detect — yet the most clinically valuable target.

Result 2: CPM & Log2 Normalization — Stabilizing the Data



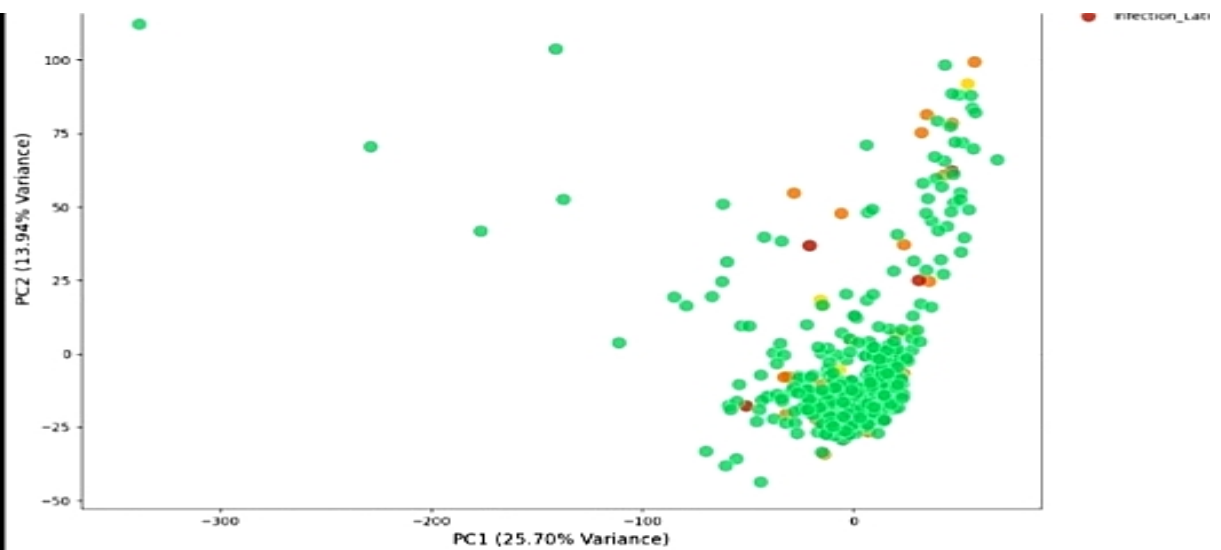
Commentary

Why CPM Normalization? Raw RNA-seq counts are biased by sequencing depth (library size). Samples sequenced more deeply appear to express every gene more highly — an artifact. CPM corrects this by scaling all samples to the same depth.

Why Log2 Transformation? Before normalization the distribution is heavily skewed — a few 'giant' genes dominate. Log2 compresses extreme values, converting multiplicative fold-changes into additive differences.

What the histogram shows: After $\text{Log}_2(\text{CPM}+1)$, the distribution follows a near-normal bell curve centred around $\text{Log}_2 \approx 7-8$. This makes T-tests and PCA statistically valid and prevents outlier genes from dominating results.

Result 3: PCA — Global Transcriptomic Separation by Infection Stage



Commentary

PC1 (25.70%): Captures the primary axis — healthy vs. infected.

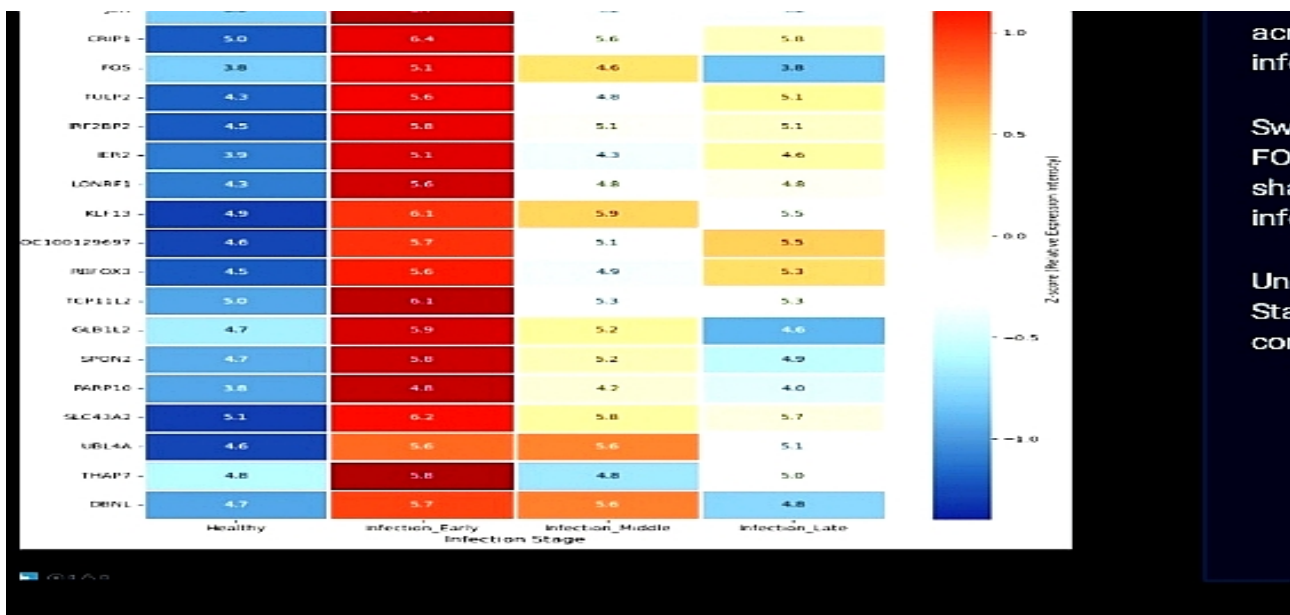
PC2 (13.94%): Captures secondary immune variation.

Healthy (green): Dense cluster on the right — stable homeostatic baseline.

Disease Trajectory: Samples shift LEFT along PC1 as infection progresses (Early → Middle).
Late-stage samples begin returning toward the healthy cluster as the immune response resolves.

Conclusion: The partial overlap between Healthy and Early stages justifies using T-test to isolate specific genes that global PCA cannot fully separate.

Result 4: Top 20 Biomarkers — Temporal Expression Heatmap



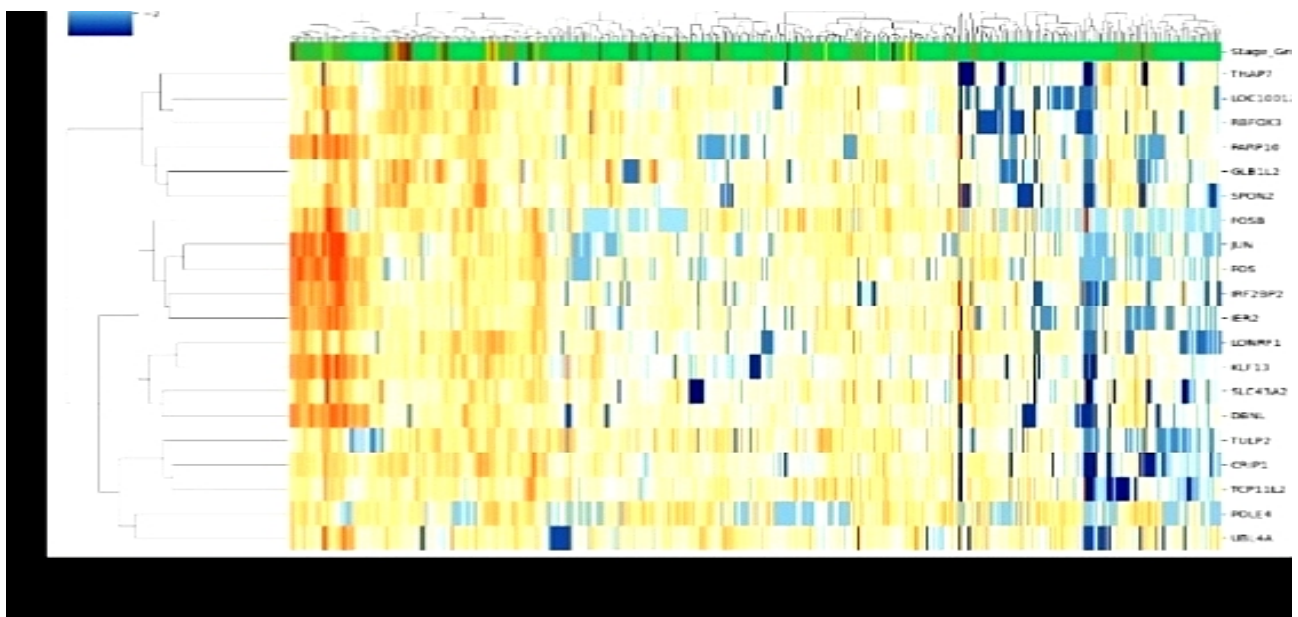
Commentary

Steady Markers — 'Always On': LOC100129697, POLE4, KLF13 maintain high expression (orange/red) across ALL infection stages. They act as constant alarms confirming infection regardless of timing.

Switching Markers — Molecular Timers: FOS and GLB1L2 surge during Early infection (deep red) then drop sharply at Late stage (blue). If FOS is HIGH → Early infection; if LOW → infection has progressed.

Unique Case — THAP7: Starts high → drops at Middle → rises again at Late. Suggests a complex biological reset during peak immune activity.

Result 5: Hierarchical Clustering — Molecular Fingerprint of Infection



Commentary

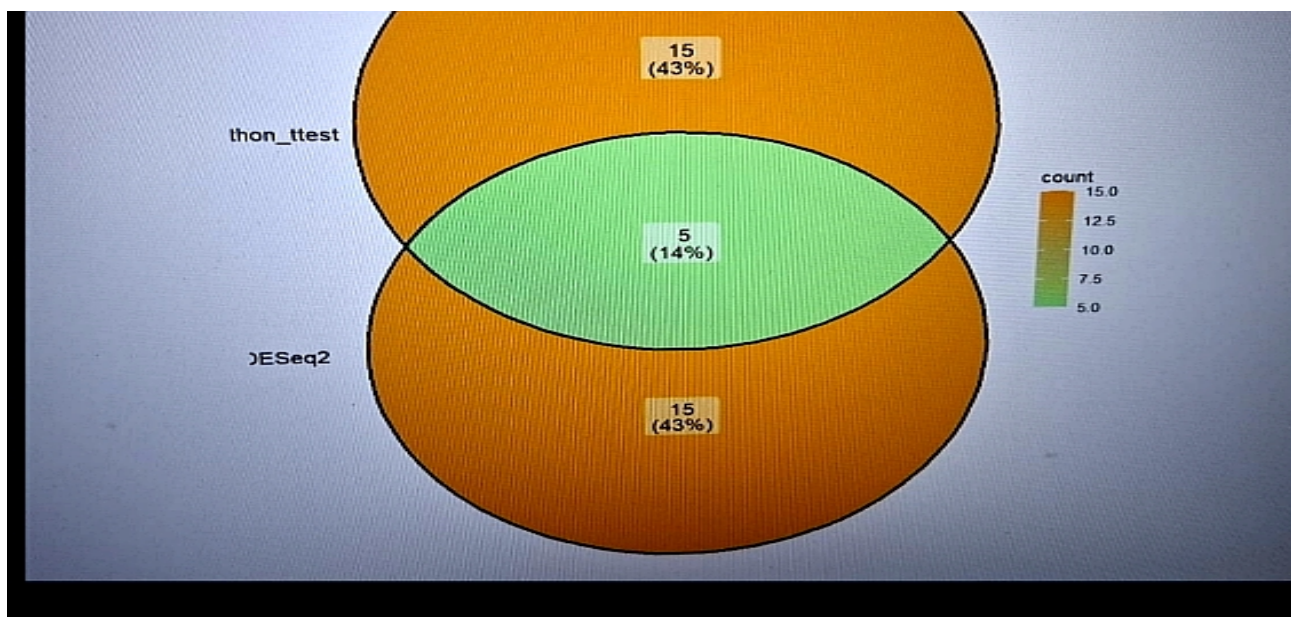
High-Resolution Patient Map: Each column = 1 patient; each row = 1 biomarker gene. The dendrogram groups similar patients and genes.

The Infected 'Red Block' (left): A clear block of high expression (red/orange) dominates the left — mostly Early & Middle patients. This is the infection's molecular fingerprint.

Healthy Cluster (right): Distinct blue-white zone — low baseline expression consistent with homeostasis.

Steady Markers: TCP11L2, POLE4, and UBL4A maintain an orange-red band across ALL infected samples — confirmed universal diagnostic anchors.

Result 6: Venn Diagram — T-test vs DESeq2 Cross-Validation



Commentary

What this shows: A comparison of differentially expressed genes identified by two independent methods: the Python T-test pipeline and DESeq2 (rigorous R-based RNA-seq analysis).

Results:

- 15 genes (43%) — T-test only
- 15 genes (43%) — DESeq2 only
- 5 genes (14%) — shared by BOTH

The 5 Shared Genes: NCAPH2, JUNB, TGFB1, DLG4, KLF13 — validated by two independent methods; highest confidence biomarker candidates.

Conclusion: T-test provides a useful first screen. DESeq2 offers more accurate dispersion modeling. The 5 shared genes are the most reliable targets for clinical validation.